

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 7 Summer 2025 (Supply.)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTMOE705A - ENGINEERING ECONOMICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

- Q1. Objective type questions. (Compulsory Question) 12
1. Which of the following is not a component of 'costs of production'?
 - a. Direct material costs
 - b. Direct labor costs
 - c. Selling and distribution overhead
 - d. Office and administrative overhead
 2. When there is decrease in the price of the product, its demand _____ and its supply _____.
 - a. decreases, increases
 - b. increases, decreases
 - c. increases, increases
 - d. decreases, decreases
 3. Direct material costs + Direct labour costs + Direct expenses = _____.
 - a. Cost of sales
 - b. Prime cost
 - c. Cost of goods sold
 - d. Sales
 4. In which stage of value engineering ideas are presented to key stakeholders?
 - a. Creative thinking
 - b. Implementation
 - c. Presentation
 - d. Evaluation

5 What are the steps in Creative thinking of value engineering?

- a. Removing redundant features
- b. Combining existing functions and components
- c. Improving build processes
- d. All of the above

In the annual equivalent method of comparison, first the annual equivalent _____ of each alternative will be computed

- a. efficiency
- b. cost or the revenue
- c. demand or supply
- d. All of the above

7 Electronic items like transistors, resistors, tubelights, bulbs, etc. could fail all of a _____, instead of gradual deterioration

- a. quarter
- b. year
- c. sudden
- d. bi-annual

8 There is a _____ between the individual replacement policy and the group replacement policy

- a. trade-off
- b. no trade-off
- c. trade-on
- d. trade-on-and-off

9 Which of the following is a Depreciation Methods?

- a. sinking fund method of depreciation
- b. declining balance method of depreciation,
- c. sum of the years digits method of depreciation,
- d. All of these

10 The ratio between the equivalent benefits and equivalent costs is known as the _____ a. maximization

- b. Benefit-Cost ratio
- c. Service Output Method of Depreciation
- d. Straight line method of depreciation,

11 In which stage of value engineering 'Evaluate ideas to develop further'?

- a. Creative thinking
- b. Implementation
- c. Presentation
- d. Evaluation

- 12 In _____, the objective is to find the single future sum (F) of the initial payment (P) made at time 0 after n periods at an interest rate i compounded every period
- Single Payment Compound Amount
 - Single-Payment Present Worth Amount
 - Equal Payment Series Sinking Fund Factor
 - Equal-Payment Series Present Worth Amount
- Q2. Solve the following.
- Illustrate the effect of price on demand and supply with the help of a diagram. 6
 - A company has to replace a present facility after 15 years at an outlay of Rs. 5,00,000. It plans to deposit an equal amount at the end of every year for the next 15 years at an interest rate of 18% compounded annually. Find the equivalent amount that must be deposited at the end of every year for the next 15 years. 6
- Q3. Solve the following.
- Explain the various steps involved in the value engineering procedure. 6
 - BYD Spare Parts company has the following details: 6
Variable cost per unit = Rs. 200; Fixed cost = Rs. 40,00,000; Selling price per unit = Rs. 400; Find (a) The break-even sales quantity, (b) The break-even sales.
- Q4. Solve Any Two of the following.
- What is the concept of 'time value of money'? Explain by considering any relevant example. 6
 - Investment proposals A and B have the net cash flows as follows: 6
- | Proposal | End of Years | | | | |
|----------|--------------|-------|-------|-------|-------|
| | 0 | 1 | 2 | 3 | 4 |
| X (Rs.) | -12,000 | 4,000 | 4,000 | 8,000 | 7,000 |
| Y (Rs.) | -12,000 | 7,000 | 7,000 | 4,000 | 4,000 |
- Compare the present worth of X with that of Y at $i = 18\%$. Which proposal should be selected? Draw the cash flow diagrams also.
- Briefly explain the 'Future Worth Method' of comparison of alternatives. 6
- Q5. Solve Any Two of the following.
- Explain the types of replacement problems. 6
 - Briefly explain the concept of challenger and defender. 6
 - A company provides a car to its chief executive. The owner of the company is concerned about the increasing cost of petrol. The cost per litre of petrol for the first year of operation is Rs.21. He feels that the cost of petrol will be increasing by Re.1 every year. His experience with his company car indicates that it averages 9 km per litre of petrol. The executive expects to drive an average of 20,000 km each year for the next four years. What is the annual equivalent cost of fuel over this period of time? If he is offered similar service with the same quality on rental basis at Rs.60,000 per year, should the owner continue to provide company car for his executive or alternatively provide a rental car to his executive? Assume $i = 18\%$. If the rental car is preferred, then the company car will find some other use within the company. 6
- Q6. Solve Any Two of the following.
- Differentiate economic analysis between the private and public sectors. 6

- B) Explain declining balance method of depreciation. 6
- C) In a particular locality of a state, the vehicle users take a roundabout route to reach certain places because of the presence of a river. This results in excessive travel time and increased fuel cost. So, the state government is planning to construct a bridge across the river. The estimated initial investment for constructing the bridge is Rs. 40,00,000. The estimated life of the bridge is 15 years. The annual operation and maintenance cost is Rs. 1,50,000. The value of fuel savings due to the construction of the bridge is Rs. 6,00,000 in the first year and it increases by Rs. 50,000 every year thereafter till the end of the life of the bridge. Check whether the project is justified based on Benefit-Cost ratio (BC-ratio) by assuming an interest rate of 12%, compounded annually. 6

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